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Real-Time Prediction of Overpressure-Induced Pipeline Failure via Edge-Optimized Machine Learning

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ABSTRACT

This study proposes a machine learning-based framework for dynamic overpressure risk prediction in gas–liquid pipelines, addressing the limitations of traditional hydraulic models in real-time operational scenarios. Unlike conventional physics-based approaches, the data-driven methodology integrates multidimensional field influence, including transient flowrates, pressure fluctuations, pigging operations, venting operations, and terrain-induced variations, to forecast overpressure-induced failure risks. Eight computationally efficient machine learning models are trained on field data and hydraulic simulation benchmarks, optimized for edge deployment with millisecond-level inference latency and minimal hardware resource consumption. The optimized algorithms demonstrate 99.6% faster computation than traditional hydraulic models while achieving satisfied prediction accuracy within 30-s warning windows, enabling real-time risk anticipation. By systematically comparing model performance across diverse operational cases, this work establishes a scalable predictive maintenance paradigm that bridges data-driven analytics with industrial safety requirements.

1 | Introduction

The pipelines undertake the most transportation tasks of fluids, including oil and gas industries, and they are one of the most potential media to transport clean energy such as hydrogen and gas hydrates as well. No matter onshore or offshore, the elevations of the pipelines are based on the terrain, which is often undulant with ups and downs. As shown in Figure 2, in gas–liquid pipelines, the gas blockages are often formed at the downward section of pipes, and are transported with the liquid phase, then they coalesce and merge into the elongated gas blockage, thus leading to the gas blockage problems including overpressure, pipe vibrations, pipe failures, and subsequent environmental issues.

Previous research has investigated the phenomenon and studied the solutions. Pothof and Clemens [1] and Chen et al. [2, 3] experimentally investigated the gas transport in the downward pipe and recorded the different forms of gas pockets under various liquid and gas velocities. Wan and Yu [4] studied the pipe-section effect on gas blockage in the undulant pipeline, especially in hump areas. Taitel et al. [5], Fabre et al. [6], Zheng et al. [7], and Al-Safran et al. [8] built the numerical models to study the principles of gas–liquid flow in hilly-terrain pipelines. Later, Chen et al. [9, 10] established the numerical models to track the gas transportation and evolution in the undulant pipeline considering the changes in gas phase morphology.

The previous work mentioned above is based on the hydraulic analysis and mathematical–physical models. These models can provide accurate hydraulic predictions in specific conditions; the models built by Chen et al. [9, 10] can simulate the gas transport accurately in the undulant pipeline in steady-state and transient-state. However, the computing cost is high for multi-case scenarios, especially for long-distance pipelines, where the simulation time will exceed several days. Meanwhile, during

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the field application, the gas transportation and other hydraulic parameters are influenced by the field factors. For instance, during the gas–liquid replacement process in the undulant pipeline, pigging processes are often launched to accelerate the gas emission. Building such processes into the model is complicated, and there are massive field factors like this, which means the traditional hydraulic models have poor adaptive capacity for field application.

Machine learning algorithms, which are widely used in the field of risk identification, have been applied in related studies. Anghel [11] established a failure-probability identifying model based on the support vector machine (SVM) algorithm. Jiang et al. [12] established the surface water extraction model in hydrological research based on the Multilayer Perceptron (MLP) algorithm and achieved the precise classification of water types. As for the energy transport industry, machine learning approaches have been applied in pipeline leak detection. Kim et al. [13] established the subsea pipeline leakage detection using an MLP approach. Banjara et al. [14] introduced an SVM approach into the pipeline leakage detection with the acoustic emission technique. These previous studies provide new insights into this work that applying various machine learning approaches to explore better solutions in different applications.

Machine learning has significantly advanced pipeline integrity management, particularly in corrosion-driven lifespan prediction. Pioneering studies demonstrate the capability of neural networks and hybrid models to achieve exceptional accuracy under operational constraints: Shaik et al. [15] brought up a feed-forward backpropagation network (FFBPN) that attains near-perfect prediction of crude oil pipeline degradation using metal loss parameters, whereas Shaik et al. [16, 17] employ Bayesian-regularized RNNs to reliably estimate pipeline lifespan even with 30% missing input data, maintaining $R^2 \approx 0.99$. The complementary work of Shaik et al. [18, 19] further validates hybrid ANN-GPR frameworks for structural health monitoring, achieving 99% accuracy in gasoline pipeline remaining life assessments. These innovations underscore the superiority of data-driven approaches over traditional physics-based models in post-failure diagnostics. However, such methods prioritize retrospective corrosion analysis over real-time risk mitigation, neglecting critical operational constraints. For instance, the computationally intensive BRNN requires GPU acceleration for training, and the ANN-GPR hybrid incurs prohibitive inference latency (second level), rendering them impractical for time-sensitive applications. Although lightweight models like SVM and RF offer faster inference (millisecond level), they fail to capture transient multiphase flow dynamics critical to overpressure risks. Furthermore, none of these studies address deployability on resource-constrained edge devices.

This study presents a novel machine learning framework for gas–liquid pipeline failure risk prediction, designed for edge computing deployment to address the latency and resource constraints. By integrating field data with hydraulic model benchmarks, we select lightweight algorithms optimized for real-time inference on low-power edge devices, enabling immediate risk analysis at the sensor source. Unlike traditional physics-based methods constrained by computational time–cost, or advanced neural network architectures dependent on intensive computational resources, our methodology bridges the gap between predictive accuracy and operational feasibility. By synthesizing transient operational dynamics with edge-optimized machine learning, the framework enables real-time anticipation of overpressure risks.

2 | The Hydraulic Prediction Model

The gas phase, which travels in the undulant pipeline in different forms, including small bubbles, elongated bubbles, and gas pockets, diminishes and even blocks the transportation of the liquid phase, and it sets a high bar for gas phase tracking and hydraulic prediction. Therefore, the hydraulic model, which is based on the work of Chen [20] by applying the gas transfer and flow pattern variation model, is introduced.

As mentioned above, the mass transfer of the gas phase is complex, as shown in Figure 1. The elongated bubble might absorb small bubbles and gas pockets, and small bubbles are entrained from the elongated one at the same time. A similar phenomenon of mass transfer can occur between gas pockets and small bubbles. Besides, mass transfer occurs between gas pockets in the form of merging, and a similar process also takes place between small bubbles through coalescence and breakup [20].

As shown in Figure 2, there are three main flow patterns in the continuous undulant pipeline: the elongated-stratified flow, bubbly flow, and slug flow. The small bubbles are entrained from the tail of the elongated bubble, which can be captured and calculated by the TBW model [21], flow with the liquid phase, and coalesce into air pockets when the bubbles' volume fraction reaches the threshold. While air pockets can be regularly formed, slug flow occurs in the pipeline [1, 3]. The migration and mass transfer between different gas forms cause the variation of flow patterns.

The small bubbles are modeled by the TBW model [21] and are tracked using the Lagrange tracking scheme, whereas the gas pockets are captured using the smooth particle hydrodynamics

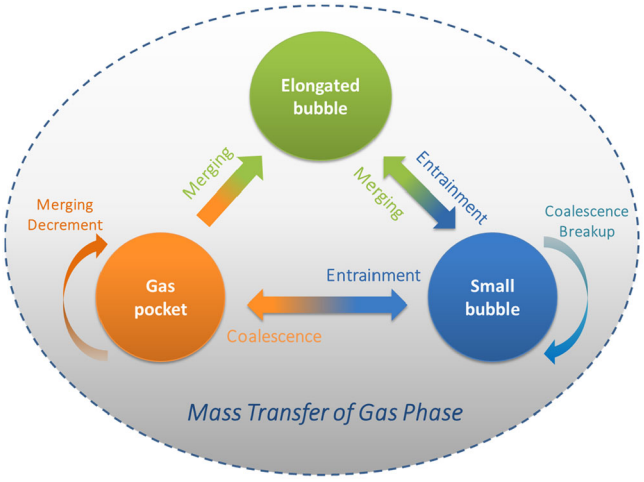


FIGURE 1 | Sketch map of mass transfer of gas phase among different forms.

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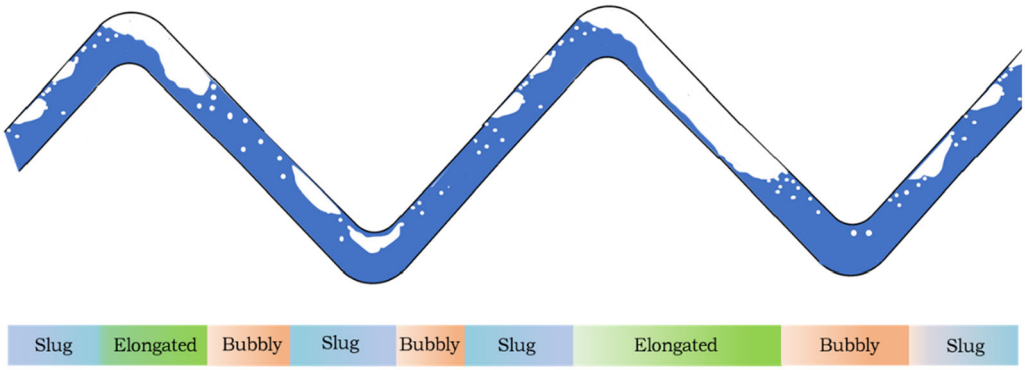


FIGURE 2 | Sketch map of flow pattern variation in the undulant pipe.

(SPH) methodology based on the small bubble track. Meanwhile, the elongated bubbles are captured and tracked according to mass and momentum conservation.

With the tracking of different gas forms in the gas–liquid undulant pipeline, the transient flow pattern map is updated at every time step, enabling pressure gradient calculation via the integrated model, as shown in Equations (1) and (2).

$$\left(-\frac{dp}{dz}\right)_{\text{in}} L_{\text{total}} = \left[\sum\left(-\frac{dp}{dz}\right)_{\text{elong}} L_{\text{elong}}\right] + \left[\sum\left(-\frac{dp}{dz}\right)_{\text{bubbly}} L_{\text{bubbly}}\right] + \left[\sum\left(-\frac{dp}{dz}\right)_{\text{slug}} L_{\text{slug}}\right] \tag{1}$$

$$L_{\text{total}} = \sum L_{\text{elong}} + \sum L_{\text{bubbly}} + \sum L_{\text{slug}} \tag{2}$$

where L_{total} indicates the total length of the pipeline, L_{elong} indicates the length of the pipe section where the elongated-stratified flow occupies, L_{bubbly} indicates the length of the pipe section where the bubbly flow occupies, and L_{slug} indicates the length of the pipe section where the slug flow occupies, the subscripts "in" is integrated, "elong" indicates elongated-stratified flow, "bubbly" indicates the bubbly flow, and "slug" is slug flow.

With this model, the relevant hydraulic parameters can be predicted in the gas–liquid pipeline; more details can be found in the work of Chen [9].

Besides, the pipe slope is another important parameter that influences the gas transportation in the undulant pipeline. However, a pipeline in the field consists of thousands of pipe slopes with various lengths; both the slope and length will influence the gas migration in the pipeline. To solve, the average slope is introduced, as shown in Equation (3).

$$S_{\text{Ave}} = \frac{\sum_{i=1}^{N} S_i \times L_i}{L_{\text{total}}} \tag{3}$$

where S_{Ave} is the average slope, S_i indicates the slope per unit length, L_i indicates the unit length of the pipe, L_{total} indicates the total length of the pipe section, and N is the total number of unit pipe sections.

3 | Machine Learning Algorithms

During the actual process of field operation, more complicated factors influence gas transport and key hydraulic parameters, including pressure gradient. For example, the pigging process is launched to push out gas blockages, and air venting valves are opened to release gas. However, these factors cannot be easily or accurately modeled in the hydraulic model. Therefore, the aim of applying machine learning algorithms is to classify severe risk levels in gas–liquid pipeline cases by considering all key factors (hydraulic and field) to support pipeline management decisions.

As shown in Figure 3, the framework of algorithm optimization and risk prediction is proposed. Pipeline data (e.g., diameter, flowrate) are included. Additionally, the average slope is considered through data processing of pipe properties, and gas blockage information derived from the hydraulic model is incorporated into risk classification. Field factors such as pigging processes and air valve venting operations, which significantly influence results, are also included.

After data processing, the data set is divided into training and test groups. Each group contains input and output data for training/testing machine learning algorithms. The trained algorithms make predictions, and assessments compare predicted outputs with test outputs to optimize algorithms for accurate risk probability calculation. To enhance reliability, eight machine learning algorithms are analyzed. These algorithms will be selected and embedded in the pipeline field edge computing devices to predict and alarm the overpressure-induced failure issues.

The selection of these eight machine learning models is driven by real-world deployment constraints in pipeline edge computing systems. Considering the millisecond-level response requirements for operational safety and the limited computational capacity of edge devices, we prioritize models with low training resource consumption, minimal inference latency, and stable execution under hardware limitations. Although advanced models (e.g., deep learning algorithms) may achieve higher accuracy in controlled environments, their excessive computational overhead and

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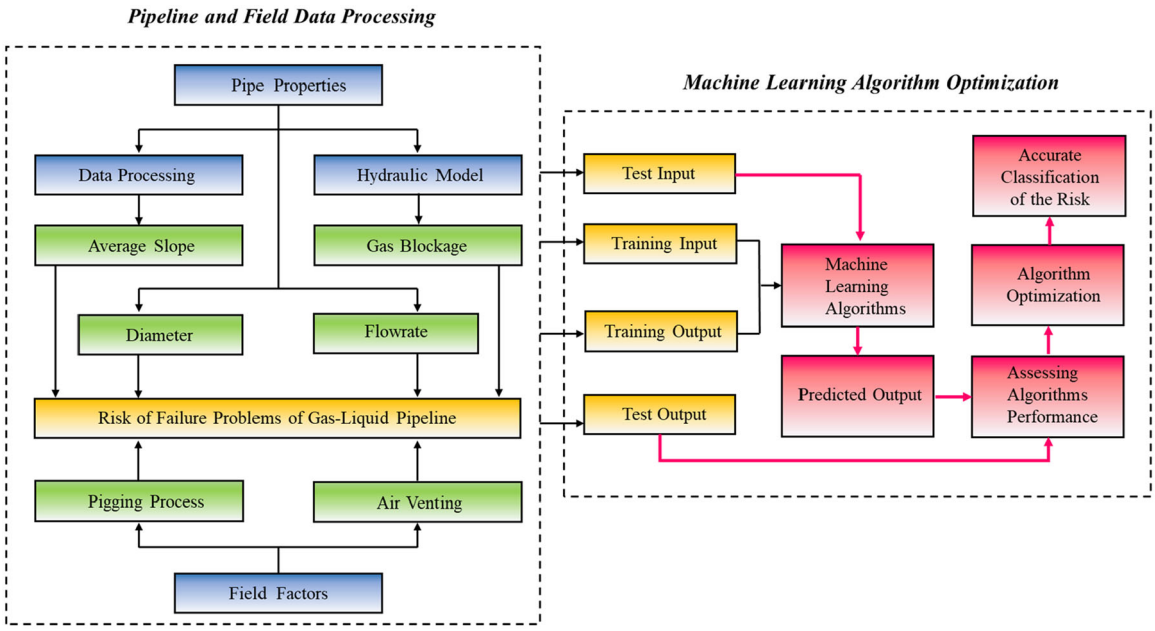


FIGURE 3 | Framework of the algorithm optimization and risk classification.

unpredictable inference times make them unsuitable for field deployment. The eight models, which are lightweight ensemble methods, are industrially require-satisfied solutions that balance prediction reliability with resource efficiency. Their mathematical simplicity also enables deterministic execution guarantees, critical for preventing catastrophic failures through real-time pressure trend prediction.

3.1 | Random Forest Algorithm

The Random Forest algorithm is an ensemble learning method based on decision trees, designed for both classification and regression tasks. It operates by constructing multiple decision trees during training and aggregating their predictions through majority voting (classification) or averaging (regression) [22]. The algorithm introduces randomness through bootstrap sampling (sampling with replacement) of the training data and random feature selection at each node split, ensuring diversity among the trees and reducing overfitting [23]. The process begins by generating N_t bootstrap samples from the original training data set. For each bootstrap sample D_t, a decision tree is constructed by recursively splitting nodes based on the Gini impurity reduction for classification tasks (Equation 4a) or variance reduction for regression. The Gini impurity at a node t is defined as

$$\text{Gini}(t) = 1 - \sum_{i=1}^{c} (p(i|t)^2) \tag{4a}$$

where $p(i|t)$ is the proportion of samples belonging to class i at node t. At each split, a random subset of m features ($m \ll$ total features) is selected to determine the optimal partition, ensuring decorrelation between trees.

The final prediction for a regression task is computed as the average of all tree predictions, as shown in Equation (4b):

$$\hat{P}_{rf}(x) = \frac{1}{N_t} \sum_{1}^{N_t} f_{N_k}(x) \tag{4b}$$

where P_{rf} is the predicted outcome from a Random Forest from a total number of N_t trees, f_{N_k} is the predictor of a single tree with an input vector x.

3.2 | K-Nearest Neighbors Algorithm

The K-Nearest Neighbors (KNN) algorithm is a nonparametric classification method that assigns a class label to a new input x based on the majority vote of its K closest training samples in the feature space [24]. The algorithm's performance heavily depends on the choice of K, which is typically optimized through cross-validation, and the selected distance metric used to quantify similarity between data points [23]. For a given input x, the algorithm first calculates the distances $d(x, x_i)$ to all training samples x_i, where the Euclidean distance is commonly defined as

$$d(x, x_i) = \sqrt{\sum_{j-1}^{m} (x^{(j)} - x_i^{(j)})^2} \tag{5a}$$

Here, m is the number of features, and $x^{(j)}$ denotes the jth feature of x. The K samples with the smallest distances are identified as the nearest neighbors.

The probability that x belongs to class k is computed as the fraction of neighbors in the K-nearest subset that belong to class k:

$$P_k(x) = \frac{N_K}{K} \tag{5b}$$

where N_k is the number of neighbors in class k. The final classification is determined by selecting the class with the highest probability.

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3.3 | Naïve-Bayes Algorithm

The Naïve-Bayes algorithm is a probabilistic classification method based on Bayes' theorem, which operates under the naïve assumption of conditional independence among input features given the class label [25]. For a given input vector $x = (x_1, x_2, ..., x_m)$, the posterior probability of belonging to class f is calculated as [23]

$$P(x) = P(Y = f|X = x) = \frac{\pi_f f_f(x)}{\sum_{l=1}^{k} \pi_l f_l(x)} \tag{6}$$

where π_f is the prior probability of randomly chosen observation and f_f is the probability density function from the fth class.

3.4 | SVM Algorithm

3.4.1 | Linear SVM Algorithm

SVM algorithms are suitable for binary classification, which is related to and contains elements of nonparametric applied statistics, neural networks, and machine learning [26]. There are two main types, one is linear SVM, another is nonlinear SVM. In the case of linear SVM, the score function is linear and parametric. The margin maximization between two data groups of solvent and insolvent is the criterion of the algorithm, where the two data groups are separated by the threshold, which is known as margin boundaries. By optimizing the margin maximization, the data set can be classified properly with acceptable and even none misclassification. The core function and constraints of Linear SVM can be shown as Equations (7a) and (7b) [26].

$$\min_{w,b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^{n} \xi_i \tag{7a}$$

$$y_i(w^T x_i + b) \geq 1\text{-}\xi_i \tag{7b}$$

where w is the weight vector, b is the bias term, ξ_i is the slack variable allowing misclassification of sample x_i, C is the penalty parameter balancing margin width and classification error.

3.4.2 | Nonlinear SVM Algorithm

The nonlinear SVM, compared to linear SVM, has a score function that is neither linear nor parametric; instead, the score of a data group is calculated by substituting the scalar product of the financial ratios with the kernel functions, which are symmetric, semi-positive definite functions satisfying the Mercer theorem [26]. The kernel functions are various and can be adjusted to the specific case, which makes the classification more accurate and reliable. The kernelized decision function of nonlinear SVM can be shown as Equation (8) [26].

$$f(x) = \text{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b \right) \tag{8}$$

where α_i is Lagrange multipliers obtained by solving the dual optimization problem, $y_i \in \{-1, +1\}$, denotes class labels of support vectors, $K(x_i, x)$ denotes Kernel function measuring similarity between x_i and x, and b is the bias term.

3.5 | MLP Algorithm

The MLP algorithm consists of one input layer, one output layer, and one or more hidden layers. Each layer contains one or more nodes connected by weighted connections, where the information transfers through forward propagation. When input data are provided, it undergoes the following forward propagation process to generate predictions: the forward propagation formula is shown as follows [27]:

$$z^{(l)} = W^{(l)} a^{(l-1)} + b^{(l-1)}, a^{(l)} = f(z^{(l)}) \tag{9a}$$

where $z^{(l)}$ is weighted sum of inputs at layer l, $W^{(l)}$ is weight matrix connecting layer $l-1$ to layer l, $a^{(l-1)}$ is activation values from layer $l-1$, $b^{(l)}$ is bias vector for layer l, and f is activation function.

The input data are processed sequentially from the input layer to the hidden layers, and the results are obtained at the output layer. If there is an error between the predicted output and the actual output, the network applies the backpropagation algorithm to adjust the weights between layers. This process repeats until the error is minimized, achieving accurate predictions. The backpropagation function is shown as follows [27]:

$$\Delta w_{ij}^{(l)} = \eta \delta_j^{(l)} a_i^{(l-1)} \tag{9b}$$

where η is learning rate $(0 < \eta < 1)$, $\delta_j^{(l)}$ is the error gradient at node j in layer l.

3.6 | Boost Algorithm

3.6.1 | AdaBoost Algorithm

The AdaBoost (Adaptive Boosting) algorithm enhances classification accuracy by iteratively combining weak learners—typically decision trees with a single split (stumps)—through dynamic sample reweighting and weighted majority voting. Initially, all training samples are assigned equal weights $w_i^{(1)} = 1/N$, where N is the total number of samples. In each iteration m, a weak learner $Gm(x)$ is trained on the weighted data set, and its weighted classification error is calculated as Equation (10a):

$$\varepsilon_m = \frac{\sum_{i=1}^{N} w_i^{(m)} I(y_i \neq G_m(x_i))}{\sum_{i=1}^{N} w_i^{(m)}} \tag{10a}$$

where $I(\cdot)$ is an indicator function (1 for misclassification, 0 otherwise). The learner's contribution to the final ensemble is determined by

$$\alpha_m = \frac{1}{2} \ln\left(\frac{1 - \varepsilon_m}{\varepsilon_m} \right) \tag{10b}$$

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which amplifies the influence of accurate learners while suppressing poor ones. Misclassified samples are then prioritized by updating their weights using Equation (10c).

$$w_i^{(m+1)} = \frac{w_i^{(m)} \exp(-\alpha_m y_i G_m(x_i))}{Z_m} \tag{10c}$$

where Z_m normalizes weights to sum to 1, and $y_i \in \{-1, +1\}$ denotes binary class labels. This iterative process forces subsequent learners to focus on challenging samples until the final ensemble the $F(x)$ in Equation (10d) achieves robust classification through aggregation of all learners' weighted predictions [28, 29].

$$F(x) = \text{sign}\left(\sum_{m=1}^{M} \alpha_m G_m(x)\right) \tag{10d}$$

3.6.2 | Gradient Boost Algorithm

Gradient Boosting is an ensemble machine learning algorithm that minimizes a differentiable loss function L by iteratively adding weak learners in the direction of the negative gradient of the loss, effectively performing gradient descent in function space [28, 30]. Unlike AdaBoost, which adjusts sample weights to prioritize misclassified examples, Gradient Boosting identifies challenging samples by computing the gradient of the loss with respect to the current model's predictions, as shown in Equation (11a).

$$r_i = -\frac{\partial L(y_i F_{m-1}(x_i))}{\partial F_{m-1}(x_i)} \tag{11a}$$

The algorithm begins with an initial model $F_0(x)$, often a constant value. At each iteration m, pseudo-residuals r_i are calculated as the negative gradient of the loss function L (Equation 11c), representing the optimal direction to reduce prediction errors. These residuals guide the training of a weak learner $h_m(x)$, which is then scaled by a learning rate v and added to the model through the update rule in Equation (11b):

$$F_m(x) = F_{m-1}(x) + v h_m(x) \tag{11b}$$

TABLE 1 | Part of data groups in the field data set.

Case no.	Flowrate (m^3/h)	Diameter (mm)	Pigging (–)	Gas-block (–)	Air-vent (–)	Ave-slope (°)	Overpressure (–)
1	890	813	No	Yes	No	4.03	Yes
2	880	813	No	No	No	4.03	No
3	900	813	No	No	No	1.26	No
4	850	813	No	No	No	1.26	No
5	880	813	Yes	No	Yes	1.26	No
6	893	813	No	No	No	1.26	No
7	855	508	No	Yes	Yes	1.16	No
8	850	508	No	No	No	1.16	No
9	851	508	Yes	Yes	Yes	1.16	Yes
10	1240	559	No	No	No	1.07	No
11	1244	559	Yes	Yes	Yes	1.07	No
12	1247	559	Yes	Yes	No	1.07	Yes
13	1246	559	No	Yes	No	1.07	Yes
14	220	406	No	Yes	Yes	3.06	No
15	224	406	No	Yes	No	3.06	Yes
16	210	406	No	Yes	No	3.06	No
17	225	406	No	Yes	Yes	3.06	No
18	199	329	No	Yes	Yes	3.06	No
19	210	329	No	Yes	Yes	3.06	Yes
20	217	329	Yes	Yes	No	2.89	Yes
21	89	219	Yes	No	Yes	4.64	No
22	88	219	Yes	No	Yes	4.64	No
23	87	219	Yes	No	Yes	4.64	No
24	93	219	No	Yes	No	4.64	No
25	95	219	Yes	No	Yes	4.64	No

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Here, ν $(0 < \nu \leq 1)$ controls the step size to prevent overfitting, whereas $h_m(x)$ approximates the residual patterns. For regression tasks, a common loss function is the squared error, as shown in Equation (11c).

$$L(y, F(x)) = \frac{1}{2}(y - F(x))^2 \tag{11c}$$

This iterative refinement continues until the loss converges or a predefined number of learners M is reached, yielding the final model $F_m(x)$.

4 | Case Study

4.1 | Data Set Information and Processing

The case study is based on a data set of 300 groups of field data. Each data group contains pipeline parameters such as diameter and average slopes of pipeline sections, hydraulic parameters such as flowrate and gas blockage, as well as field parameters including pigging processes and air venting processes. These parameters are key factors profoundly influencing flow assurance in undulant gas–liquid pipelines; thus, they are selected to predict severe pipeline risks. To enhance the model's applicability, data are extracted from diverse undulant pipelines with varying parameters and operational conditions, ensuring hydraulic and field parameters are not held constant.

As shown in Table 1, a subset of the field data set is presented. Seven key parameters are recorded; some are binary, whereas others are numerical. In each case, data are collected from pipe sections whose length varies by application. Specifically, data are collected from sections between pump stations. Feature parameters include flowrate, diameter, pigging status, gas blockage, air venting status, and average slope, with the label parameter being overpressure failure occurrence. The table shows data from pipelines with different diameters, flowrates, slopes, and field factors such as pigging process activation and air venting valve placement. The data set comprises 300 groups, all used for model training and testing.

Meanwhile, to avoid the imbalance problems of the data set, synthetic minority over-sampling technique (SMOTE) is employed. The percentage of high-risk data groups is relatively low among the data set, because the field pipelines are all predesigned by the feasibility analysis; the pipelines are supposed to work well, but exceptions appear when the gas blockages are severe and cause severe pipeline problems. Therefore, the cases where severe problems occur share a relatively low portion in the data set. Such an imbalance could increase the risk of misclassification of the model. To solve this, the SMOTE is applied. The technique optimizes the data set by creating synthetic minority class data to achieve the over-sampling of the minority data group, so the imbalance problem can be resolved as proposed by Chawla et al. [31].

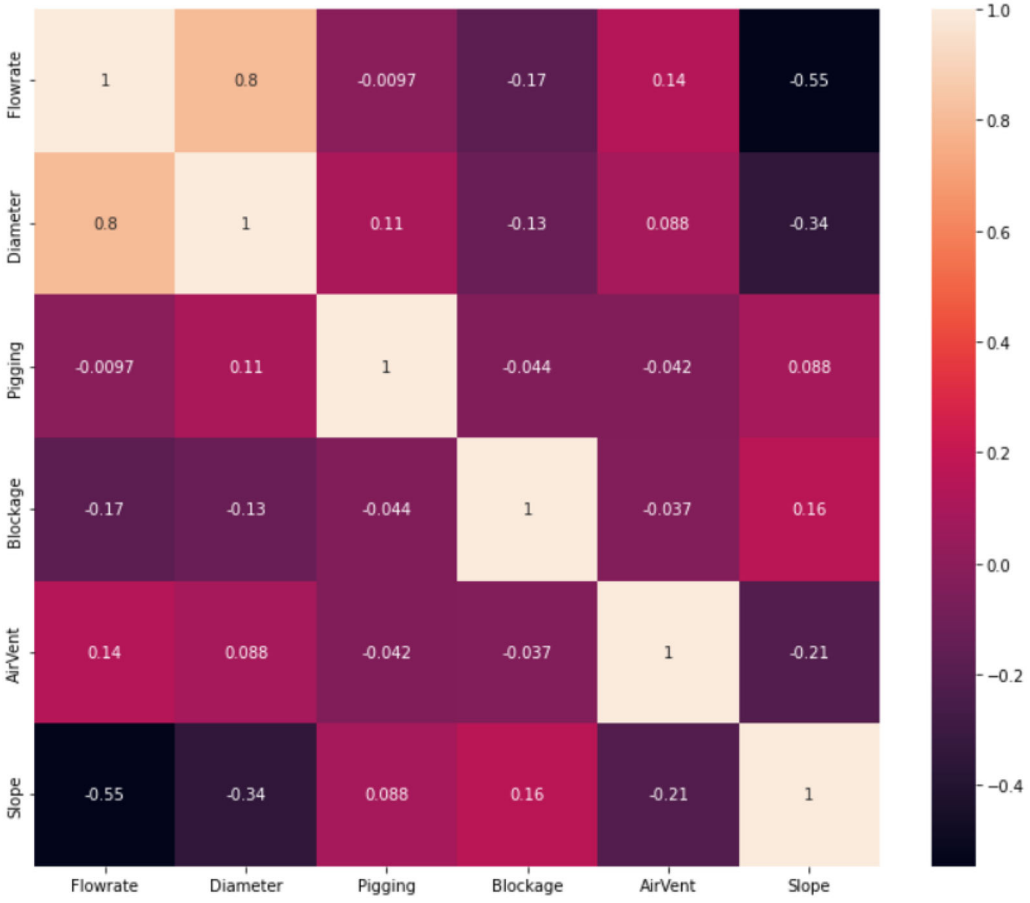


FIGURE 4 | Correlation matrix for classification.

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4.2 | Model Performance

After the data set processing, a correlation matrix is generated to learn the relationship between different types of input data, and it is shown in Figure 4. Those input pairs that have a strong correlation with each other will gain high value in the corresponding cell. From the correlation matrix, it can be shown that there are strong bonds between some of the input pairs while other pairs are weakly bonded. Positive value in the cell represents that the input pair is positive-related, whereas the negative value means the input pairs are negative-related, which means one's increase will result in another's decrease. In Figure 4, diameter and flowrate are highly positive-bonded, because the larger the diameter, the higher flowrate that the pipeline is designed to undertake. Oppositely, flowrate and diameter are negatively bonded with the average slope; that might be the result of the safety consideration of design, those pipelines with large diameters and high flowrates would not take risk crossing the steep terrain.

In Figure 5, the feature importance of the Random Forest algorithm, AdaBoost algorithm, and Gradient Boost algorithm is revealed. The features in these algorithms provide the estimates of what variables are important in the classification, and they help the algorithms handle thousands of input variables without variable deletion. The feature importance can be measured as the average impurity decrease computed from all

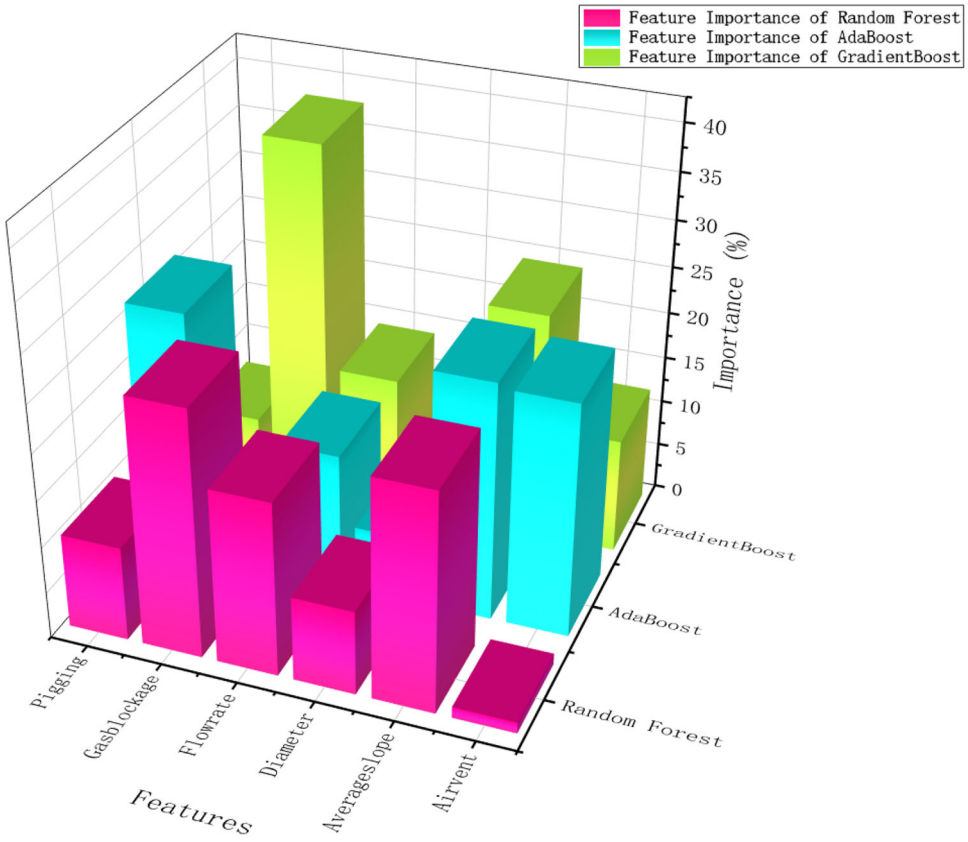


FIGURE 5 | Feature importance.

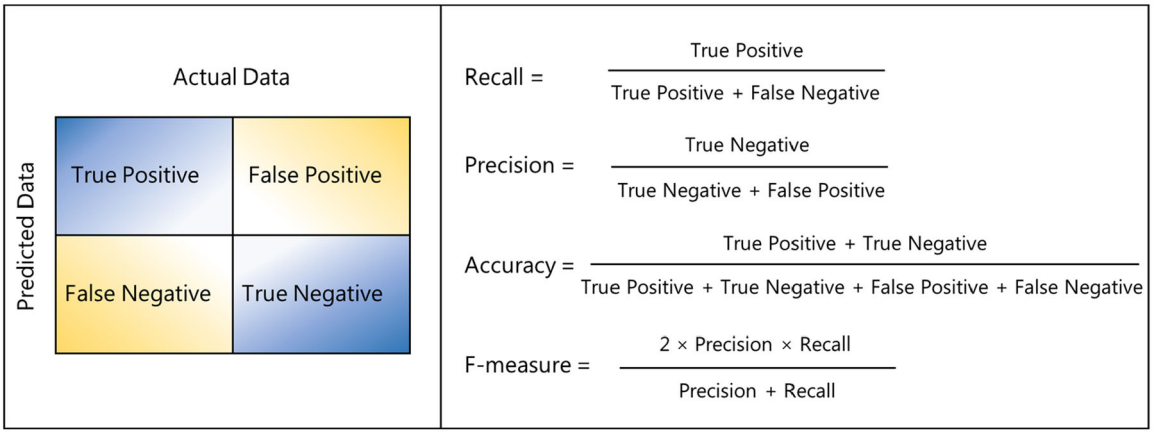


FIGURE 6 | Confusion matrix and performance metrics.

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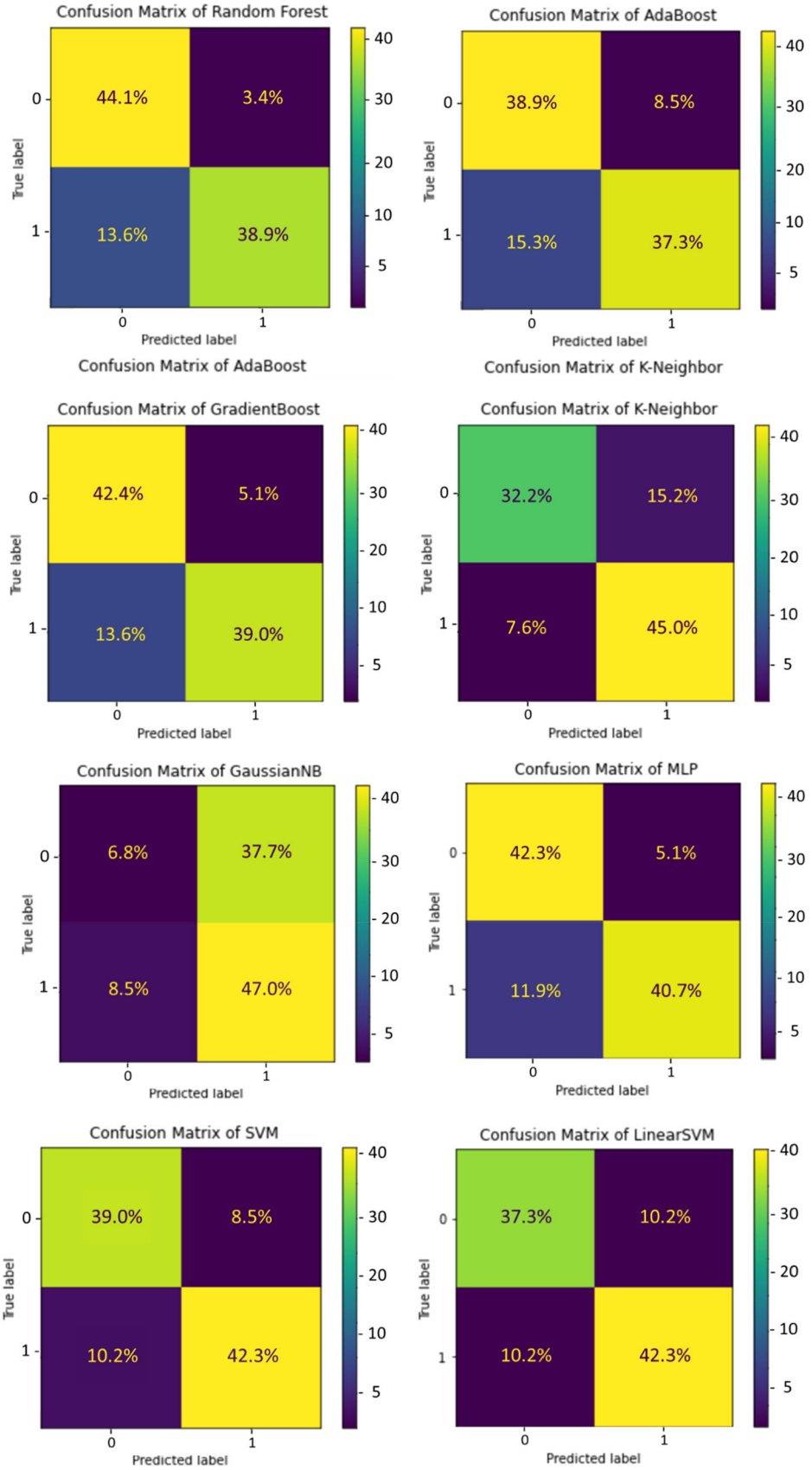


FIGURE 7 | Confusion matrix of different algorithms.

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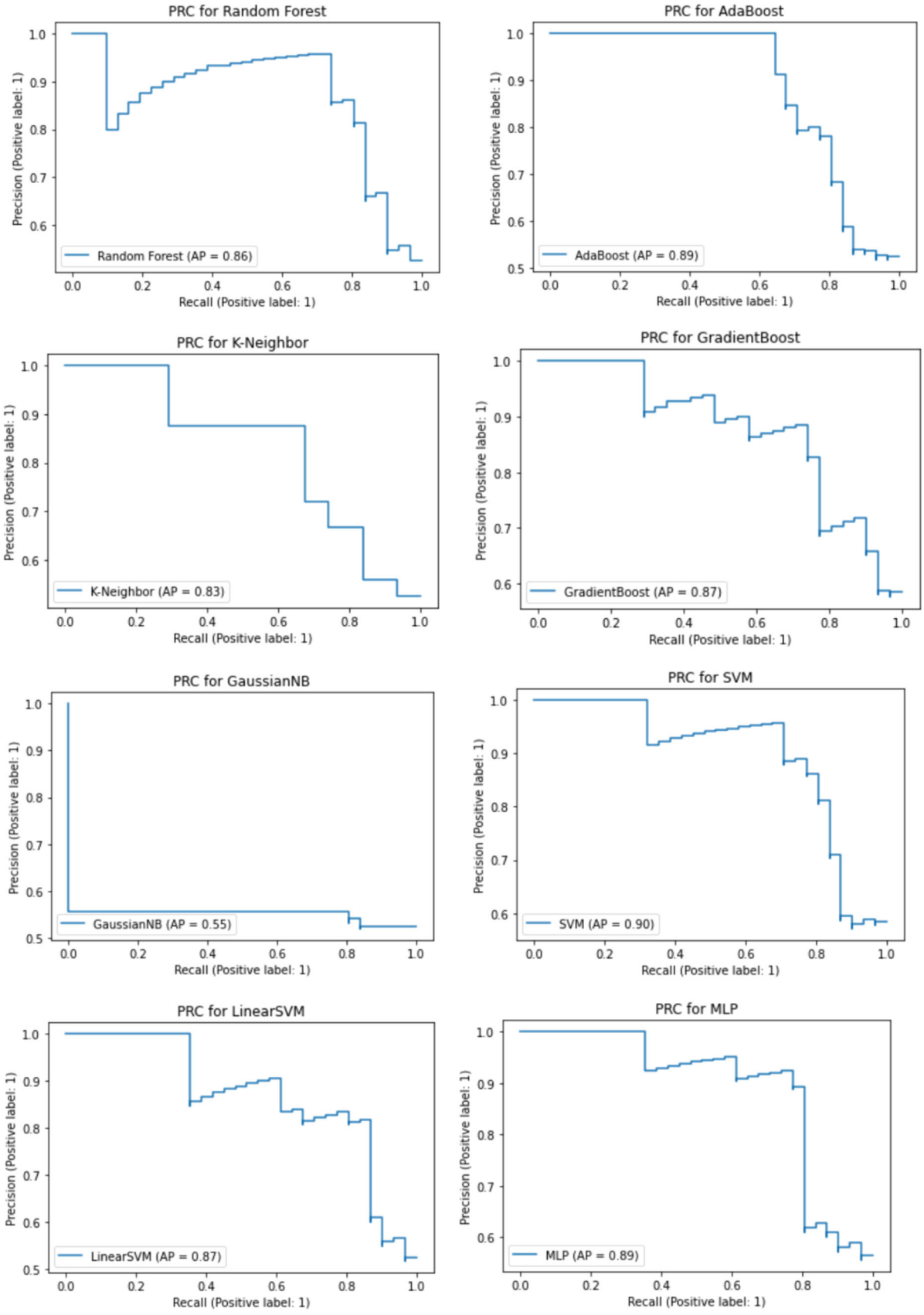


FIGURE 8 | Precision–recall curves of different algorithms.

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decision trees, or stumps in the forest. This is irrespective of the fact whether the data are linear or nonlinear. In Figure 5, gas blockage with the importance of 38.2% and average slope with 24.5% are the most important features in the Gradient Boost algorithm, whereas in the AdaBoost algorithm, air-vent, average slope, and pigging share the most importance around 26%. The gas blockage 26.5% and average slope 23.5% have the most influence in the Random Forest algorithm.

4.2.1 | Confusion Matrix

The confusion matrix is a technique for summarizing the performance of a classification algorithm. Classification accuracy alone can be misleading if there is an unequal number of observations in each class or if there are more than two classes in the data set. A better understanding of classification models can be gained by calculating the confusion matrix, which identifies which predictions are correct and which are not. In other words, the number of correct and incorrect predictions is summarized with count values and broken down by each class.

As shown in Figure 6, True Positive and True Negative represent the correct predictions of the model, whereas the False Positive and False Negative show the inaccurate predictions. Besides, such a matrix will provide us more performance metrics of the prediction assessment. The recall and precision of the prediction, as well as accuracy and F-measure, can be calculated based on the confusion matrix and thus to evaluate the prediction in a more comprehensive way. The confusion matrix of various classification algorithms is presented in Figure 7.

4.2.2 | Precision and Recall

Based on the confusion matrix, the recall and precision of the predictions will be analyzed. Rather than simply listing and

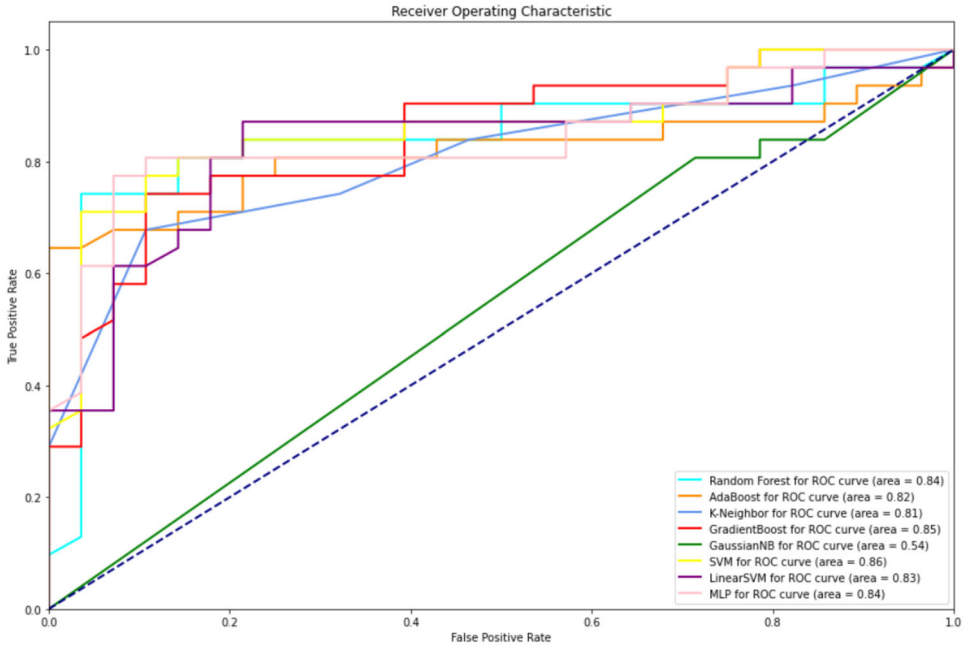


FIGURE 9 | ROC curves and AUC values of various algorithms.

TABLE 2 | Overall assessment of various algorithms' performance.

	Accuracy		AP		F-measure		AUC	
	Score	Rank	Score	Rank	Score	Rank	Score	Rank
Random Forest	0.83	1	0.86	6	0.83	1	0.84	3
AdaBoost	0.76	6	0.89	2	0.77	6	0.82	6
K-Neighbor	0.71	7	0.83	7	0.70	7	0.81	7
Gradient Boost	0.81	3	0.87	4	0.82	3	0.85	2
Naïve-Bayes	0.51	8	0.55	8	0.43	8	0.54	8
Nonlinear SVM	0.81	3	0.9	1	0.82	3	0.86	1
Linear SVM	0.79	5	0.87	4	0.8	5	0.83	5
Multilayer Perceptron	0.83	1	0.89	2	0.84	1	0.84	3

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comparing the values, the precision–recall curve (PRC), which shows the relationship between precision and recall, is employed to evaluate the prediction models. The PRC can be the supplementary evaluation approach regards to the receiver operating characteristics (ROC) and area under the curve (AUC). For instance, if the classifier tends to find anomaly detection when the data are heavily imbalanced in the negative class, it will gain poor classification performance even if the AUC is superior [32]. With applying the PRC in such cases, the classifier performance evaluation will be more accurate. As shown in Figure 8, the *x*-axis represents the recall, whereas the *y*-axis represents the precision; the PRC of various algorithms is presented. Meanwhile, the important evaluation, the average precision (AP), which is the area under the PRC, can be calculated and analyzed.

In Figure 8, the PRC and AP of various algorithms are presented. It can be shown that the various PRCs can be zigzag and make it difficult to compare, but it can be persuasive if the comparison is processed with AP. From Figure 8, it can be seen that the SVM algorithm achieves the most superior AP of 0.90, followed by the MLP algorithm (AP = 0.89) and AdaBoost algorithm (AP = 0.89).

4.2.3 | Accuracy and F-Measure

As shown in Figure 6, the accuracy and F-measure can be used for the evaluation of classification performance as well. The accuracy is used to calculate the accuracy of either the faction or count of correct predictions, and mathematically it represents the ratio of the sum of true positives and true negatives out of all the predictions. The F-measure can be interpreted as a harmonic mean of the precision and recall, where an F-measure reaches its best value at 1 and worst score at 0. The relative contribution of precision and recall to the F-measure is equal.

4.2.4 | ROC and AUC

The ROC curve, which is based on the Neyman–Pearson method, has been widely used to assess the classifier's performance. It shows the relationship between True Positive and False Positive. Besides, the AUC is a meaningful interpretation for the classifier assessment [33]. Meanwhile, the AUC is not dependent on a decision threshold chosen when compared to overall accuracy, and it gives an indication of the extent that negative and positive classes are separated [34]. Therefore, the ROC and AUC can be an ideal tool to assess the performance of classifiers.

As shown in Figure 9, the *x*-axis represents the False Positive rate, whereas the *y*-axis represents the True Positive rate. All the algorithms except Gaussian Naïve-Bayes show good performance in the prediction, and those ROC curves intersect with each other to reach their best performances. Therefore, a convincing way to assess is to compare the value of AUC. The leading algorithm of SVM achieves the AUC of 0.86, followed by Gradient Boost (AUC = 0.85), MLP (AUC = 0.84), and Random Forest (AUC = 0.84).

4.3 | Model Analysis and Discussion

As Table 2 shows, four assessment aspects are considered, including accuracy, AP, F-measure, and AUC (area under the curve of ROC). Every single performance of algorithms regarding these aspects is scored and ranked in the table, which is intuitively presented. The assessment can be made via overall-considering the ranks, and it is obvious that the Random Forest algorithm and MLP algorithm achieve the best ranking, which means both of them have satisfactory performances in the severe risk prediction of the undulant gas–liquid pipeline.

With applying and optimizing the machine learning algorithms to the risk prediction of failure problems during the gas–liquid pipeline, the cost of computing dramatically decreases. A new data set consisting of 100 groups of field data, which is different from the previous one mentioned before, is included in this analysis. Note that this analysis is based on the same computational capacity.

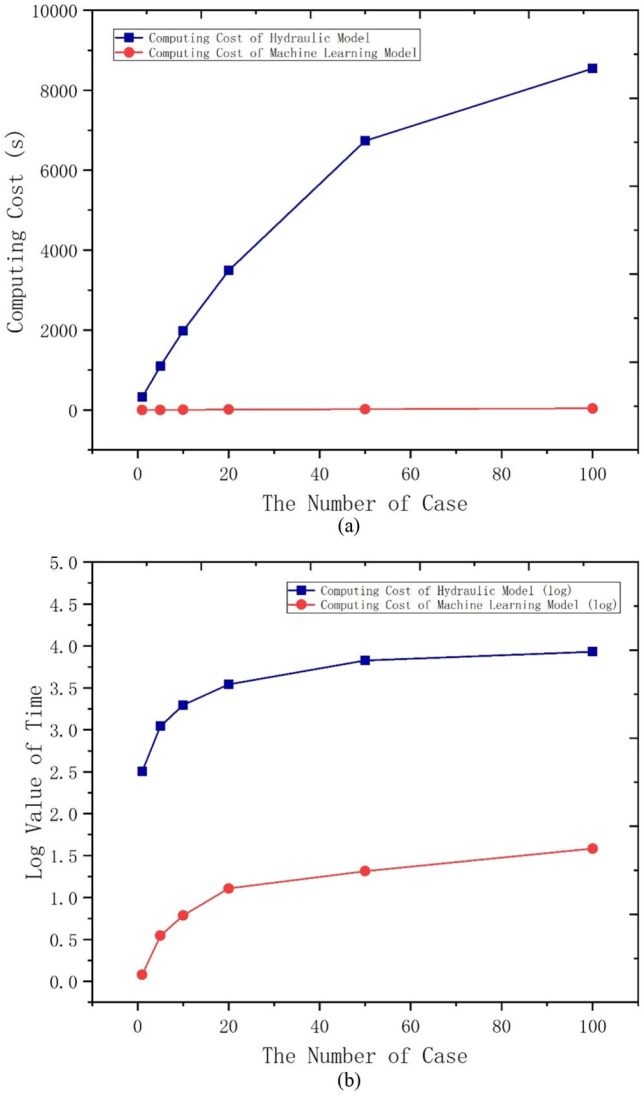


FIGURE 10 | Computing cost comparison between the hydraulic model and the machine learning model: (a) original data value and (b) logarithmic data value.

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As shown in Figure 10a, the machine learning model achieves about 100 times faster calculation than the hydraulic model in a small number of cases. As the number of cases increases, the computation cost gap becomes more obvious, eventually reaching 3400 times at the 100-case calculation. To gain a more accurate observation of the red line in Figure 10a, the logarithmic axis is introduced, and the result is shown in Figure 10b. It is obvious that both costs of the models increase with the case number. In summary, the machine learning model is more advantageous and time-saving for multiple cases.

Another crucial comparison is the accuracy of the calculation. The field factors are not included in the original hydraulic model; therefore, the first accuracy comparison is processed without field factors. As shown in Figure 11, the accuracy of the machine learning model is over 80%, which is acceptable in many scenarios, especially when the number of cases is large. Far more time is required for hydraulic models.

4.4 | Pipeline Field Application

The implementation of machine learning algorithms in pipeline edge computing systems requires rigorous integration with field-grade hardware to achieve real-time overpressure risk prediction. The edge computing devices deployed in this application, such as industrial controllers based on Raspberry Pi Compute Module 4 architectures, are equipped with quad-core ARM Cortex-A72 CPUs operating at 1.8–2.0 GHz, 4 GB LPDDR4 RAM, and 64 GB eMMC storage. This type of device is engineered to withstand harsh environmental conditions, including ambient temperatures ranging from −40°C to +85°C, IP67-rated enclosures for protection against dust and moisture ingress, and ultra-low power consumption below 15 W to support solar or battery-powered operations in remote pipeline segments, and affordable cost of $200–300 per unit.

The eight machine learning models, including Random Forests, SVM, and MLP, are optimized to align with the computational limitations of these edge devices. During inference, the models maintain a memory footprint under 500 MB and utilize less than the CPU core capacity per prediction cycle. This efficiency ensures seamless coexistence with essential SCADA tasks such as TCP data communication, local human–machine interface rendering, and real-time actuator control. Key performance benchmarks validate their operational viability: Each inference is completed at millisecond-level latency, and each device concurrently monitors 2–4 pipeline segments without performance degradation.

As shown in Figure 12, these algorithms process 1 Hz telemetry streams capturing dynamic parameters from pump stations, like pressure fluctuations, flowrate variations, pigging operation statuses, and pipe venting operations. The output comprises two actionable metrics: an instantaneous overpressure probability, updated every second, and a least 30-s risk escalation forecast derived from trend analysis. The distinction between real-time overpressure probability assessment and 30-s-ahead predictive analytics lies in their inherent differences in temporal granularity and risk mitigation paradigms. Real-time probability quantification evaluates instantaneous system states to determine the immediate risk level, serving as a continuous

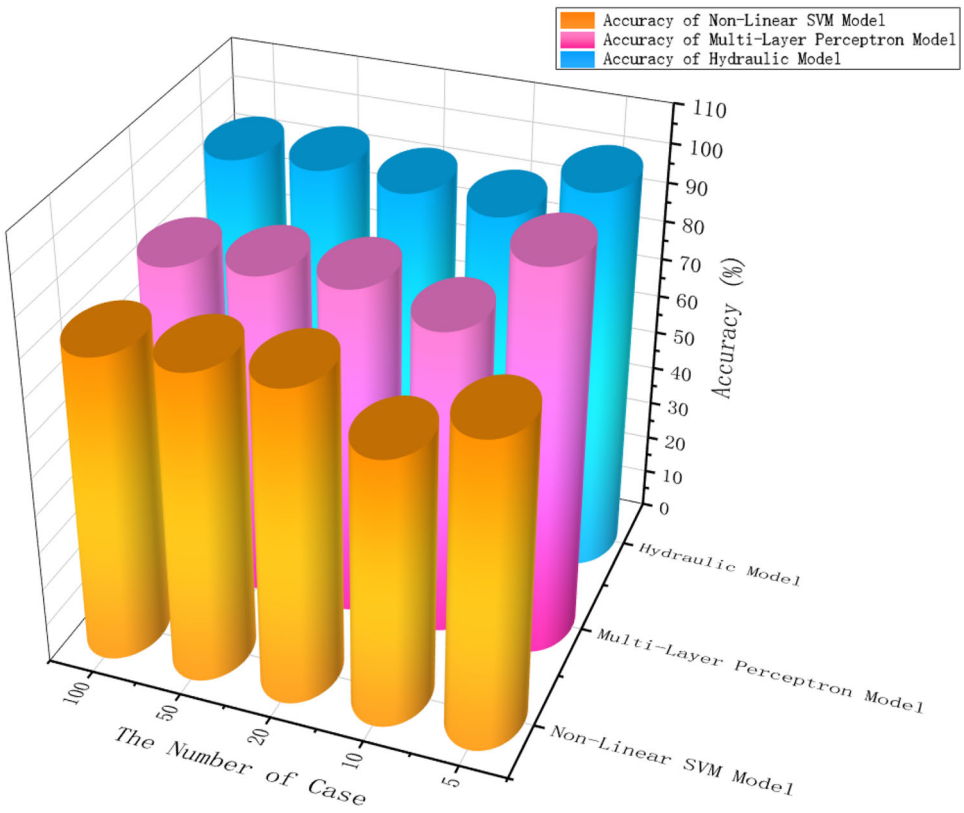


FIGURE 11 | Accuracy comparison among the hydraulic model and well-performed machine learning models.

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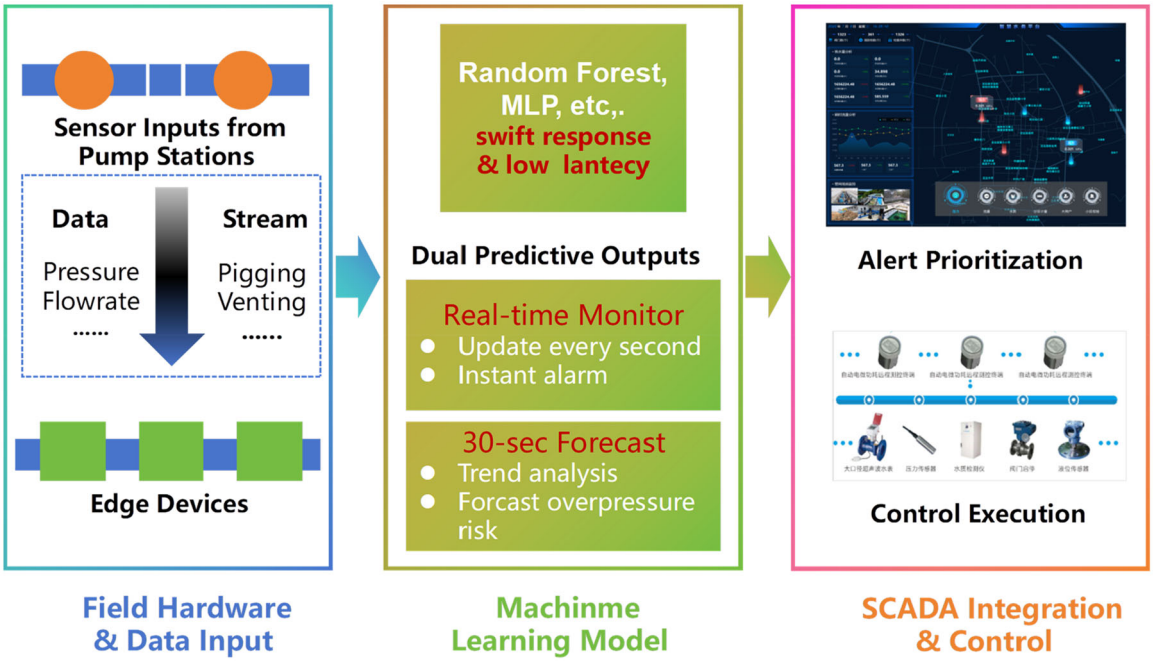


FIGURE 12 | Pipeline field application workflow.

monitoring mechanism for early anomaly detection. In contrast, 30-s-ahead prediction models forecast cumulative risk escalation by simulating dynamic parameter interactions over time windows, such as pressure surge trajectories induced by pigging operations or severe gas blockage formation, thereby anticipating threshold breaches before physical manifestations occur. While the former prioritizes sensitivity to transient operational fluctuations, the latter emphasizes trend-driven intervention planning. These dual predictive layers synergistically enhance pipeline safety: Real-time probabilities trigger vigilance alerts including operator notifications for emerging risks, whereas forward-looking forecasts enable preemptive control actions such as pre-venting valve activation to avert irreversible failures. This hierarchical framework bridges short-term responsiveness with medium-term prognostics, ensuring operational stability under both steady-state and transient conditions.

Alerts are transmitted to SCADA interfaces, where operators or smart systems receive prioritized recommendations such as reducing compressor output by 10% within 20 s. These commands are directly integrated into programmable logic controller systems, achieving end-to-end latency from sensor input to control signal execution, thereby enabling proactive risk mitigation in critical infrastructure.

5 | Conclusions

This paper established machine learning models, driven by 300 field data sets and key hydraulic simulation results, to achieve accurate failure prediction in gas–liquid pipelines. Eight algorithms, including nonlinear SVM, Random Forest, Boosting methods, Naïve-Bayes, KNN, and MLP, were rigorously optimized and compared. The models not only complement traditional hydraulic approaches but also demonstrate significant advantages in computational efficiency and adaptability to edge computing environments for real-time pipeline monitoring. Key conclusions are summarized as follows:

i. *Enhanced field insights and adaptability*. The machine learning algorithms identified critical field factors that were not explicitly modeled in the original hydraulic framework. These insights improve risk prediction accuracy and can be generalized to diverse pipeline configurations. By integrating lightweight model retraining on edge devices, the framework adapts to new pipeline geometries with minimal accuracy loss, enabling rapid deployment in remote or resource-constrained environments.

ii. *Algorithm performance*. The Random Forest and MLP algorithms achieve the best performance among the eight algorithms during the risk prediction of failure problems in the gas–liquid pipeline.

iii. *Computational efficiency and feasibility*. The machine learning models dramatically reduced computational costs compared to hydraulic simulations, achieving a 3400-time speedup in processing 100 cases. The average accuracy of machine learning models is acceptable in many occasions especially when the case number is large. Therefore, when it comes to larger amounts of cases, the machine learning models' advantages will be obvious.

Data Availability Statement

The data that supports the findings of this study are available within the article.

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